

Query Representation and Processing

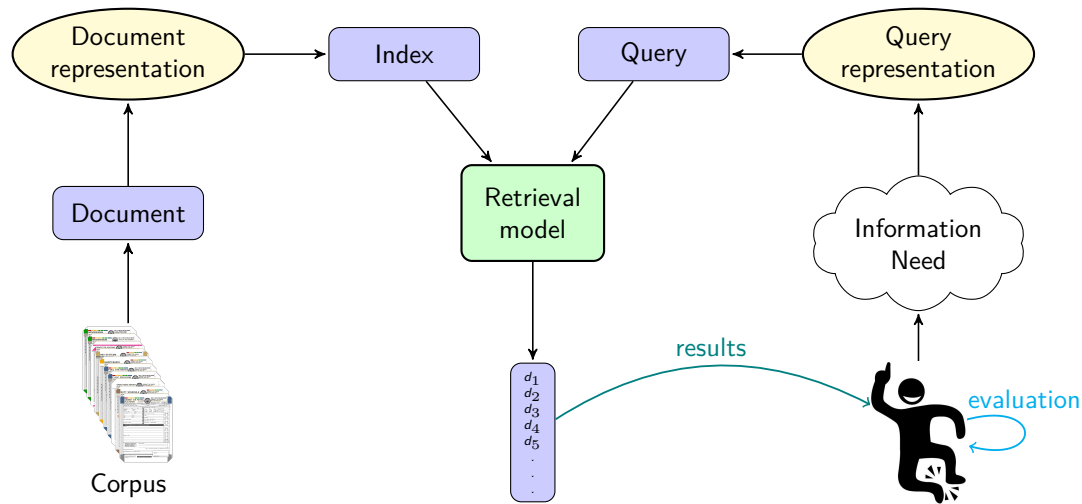
Introduction to Information Retrieval

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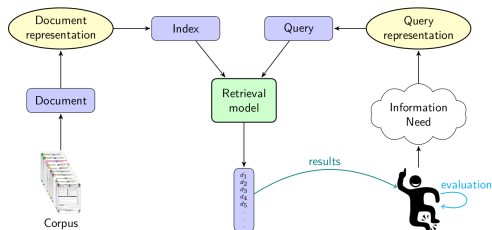
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Information Retrieval - A Graphical Representation



Information Retrieval - A Graphical Representation



- ✓ **Document representation:** determines what goes in the index (e.g., term occurrences, term frequencies, etc.).
- ✓ **Index:** facilitates quickly finding the documents that match the query.
- ➔ **Query representation:** defines how different types of queries can be structured before retrieval can be performed.
- **Retrieval model:** decides whether a document is relevant to the query (and possibly its degree of relevance).

Binary full-text representation

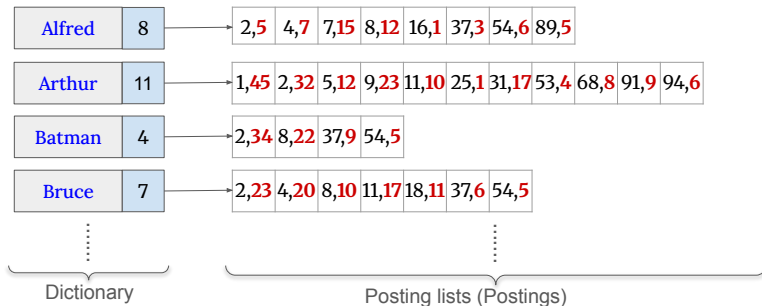
boolean retrieval

	Inception	The Dark Knight	Interstellar	Dunkirk	The Prestige	Memento
Emma Thomas	1	1	1	1	1	1
Cillian Murphy	1	1	0	1	0	0
Michael Caine	1	1	1	0	1	0
Tom Hardy	1	0	0	1	0	0
Christian Bale	0	1	0	0	1	0
Guy Pearce	0	0	0	0	0	1
Hugh Jackman	0	0	0	0	1	0

- Bag of words representation (no frequency, location information stored).
- Facilitates unranked boolean retrieval.
- Facilitates boolean operators **AND**, **OR**, and **AND NOT**.
- Facilitates efficient query processing:
 - ▶ computers can execute boolean operations fast (e.g., 10110011 **AND** 01001111 = 00000011)

Full-text Representation

variable-length inverted lists with term-frequencies



- Bag of words representation (no term-location information)
- Facilitates unranked and ranked boolean retrieval
- Facilitates boolean operators **AND**, **OR**, and **AND NOT**
- Facilitates efficient query processing
 - ▶ merging inverted lists can be done quickly

What about other query operators?

- A boolean query language with only **AND**, **OR** and **AND NOT** is too restrictive.
- Difficult to balance precision and recall.

What about other query operators?

- A boolean query language with only **AND**, **OR** and **AND NOT** is too restrictive.
- Difficult to balance precision and recall.
- Systems that employ boolean retrieval include other operators:
 - ① **proximity operators**: impose constraints on query-terms appearing close to each other (*ordered* or *unordered*).
 - ② **field restriction**: impose constraints on terms appearing in particular parts of the document.
 - ③ **wild-card operators**: impose constraints on matching query-terms with index-terms.

Proximity Operators

ordered window (OW)

- **A** OW/**N** **B**: **A** must appear no more than **N** terms *before* **B**.

Proximity Operators

ordered window (OW)

- **A** OW/**N** **B**: **A** must appear no more than **N** terms *before* **B**.
- Example: (information OW/2 science)
 - ✓ information library science
 - ✓ information science
 - ✗ information and library science
 - ✗ science of information

Proximity Operators

phrase operator

- **A B**: **A** must appear immediately before **B**
- Example: information science
 - ✗ information library science
 - ✓ information science
 - ✗ information and library science
 - ✗ science of information
- equivalent to (information OW/1 science)

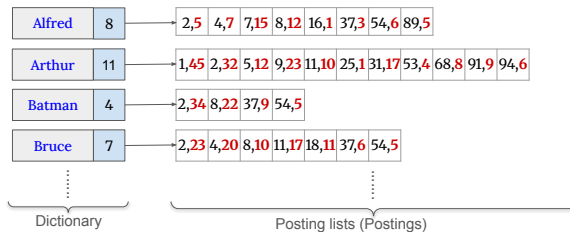
Proximity Operators

unordered window

- **A** **UW/N** **B**: **A** must appear no more than **N** terms before or after **B**
- Example: (**information UW/2 science**)
 - ✓ **information library science**
 - ✓ **information science**
 - ✗ **information and library science**
 - ✓ **science of information**

Full-text representation

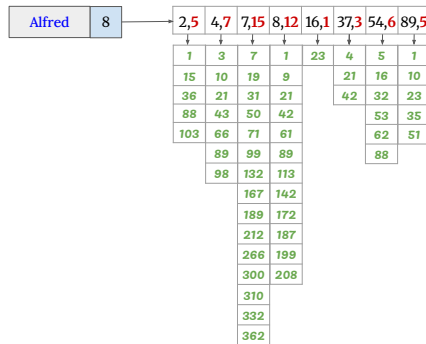
variable-length inverted lists with term frequencies



- We cannot handle proximity operators with this index.
- ? What additional information do we need?

Full-text representation

positional index



- docid, tf [pos₁, pos₂, ..., pos_{tf}]
- ✓ Which document(s) match **information library science**?
- ✓ What about (**information OW/2 science**)?

Relevance-Based Language Models

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ABSTRACT

We explore the relation between classical probabilistic models of information retrieval and the emerging language modeling approaches. It has long been recognized that the primary obstacle to effective performance of classical models is the need to estimate a relevance model: probabilities of words in the relevant class. We propose a novel technique for estimating these probabilities using the query alone. We demonstrate that our technique can produce highly accurate relevance models, addressing important notions of synonymy and polysemy. Our experiments show relevance models outperforming baseline language modeling systems on TREC retrieval and TDT tracking tasks. The main contribution of this work is an effective formal method for estimating a relevance model with no training data.

1. INTRODUCTION

Recent developments in the conceptual view of Information Retrieval marked a departure from the traditional models of relevance and the emergence of language modeling frameworks, introduced by Ponte and Croft [16]. A number of following publications [5, 14, 21, 10], adopted a similar framework, introducing refinements in parameter estimation, the use of multi-word features and expansion techniques. A common theme in these models is that they abandon explicit models of relevance, instead attempting to model the query generation process. These approaches model query generation as random sampling from one of the document models. During the retrieval process, the documents are ranked by the probability that a query would be observed as a random sample from the respective document model.

Earlier work on probabilistic models of information retrieval [19, 18, 17, 22] took a conceptually different approach. Researchers explicitly attempted to model word occurrences in relevant and non-relevant classes of documents, and used their models to classify the document into the more likely class. For retrieval purposes, the documents were ranked by the probability that they belong to the relevant class.

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The difference between the two approaches is not superficial, but is in fact quite important. Traditional probabilistic frameworks allow for arbitrarily rich and complex query models (e.g., [22]), whereas the recent language modeling approaches treat the query as a fixed sample of text with little room for modeling. Many popular techniques in Information Retrieval, such as relevance feedback and automatic query expansion have a very intuitive interpretation in the traditional probabilistic models, but are conceptually difficult to integrate into the language modeling framework of [16] because they involve augmenting the sample (as in [15]) rather than adjusting the model. Furthermore, explicit models of relevance are better suited to other information organization tasks, such as summarization [6], question answering [4], topic detection and tracking (TDT) [25, 12] and text segmentation [3].

The primary obstacle to constructing effective models of relevance is the lack of training data. In a typical retrieval environment we are given a query, a large collection of documents and no indication of which documents might be relevant. In this paper we introduce a technique for constructing a relevance model from the query alone and compare resulting query-based relevance models to language models constructed with training data. We also evaluate the retrieval effectiveness of query-based relevance models on two distinct tasks: the *ad-hoc* retrieval task of TREC [1], and the tracking task of TDT [2]. The remainder of the paper is structured as follows. We briefly survey the classical relevance-based probabilistic models as well as the recent language modeling approaches in section 2. Our method for formally constructing and applying relevance models is detailed in section 3. Section 4 provides a thorough evaluation of query-based relevance models in terms of cross-entropy, retrieval effectiveness and tracking accuracy.

2. RELATED WORK

There are two directions of information retrieval research that provide a theoretical foundation for our model: the now classic work on probabilistic models of relevance, and the recent developments in language modeling techniques for IR. To the former we owe the concept of a relevance model: a language model representative of a class of relevant documents. To the latter we credit the methods of modeling and specific estimation techniques. In this section we give a brief survey of several developments in both of these directions, highlighting interesting connections between the two.

2.1 Classical Probabilistic Models

Underlying most research on probabilistic models of Information Retrieval is the probability matching principle, detailed by Robertson in [19], which suggests ranking the documents D by the odds of their being observed in the relevant class: $P(L|R)/P(L|N)$.

Field-restricted Search

indexing

- Build a separate index for each field type



Field-restricted boolean retrieval

query processing

- Process each field-specific subquery using the appropriate index and merge the results (as usual)

Field-restricted boolean retrieval

query processing

- Process each field-specific subquery using the appropriate index and merge the results (as usual)
- We may also provide **boosting** to some fields.

Wildcard operators

- Some query languages allow users to match a single query-term to multiple index-terms
- **light AND therap***

Wildcard operators

- Some query languages allow users to match a single query-term to multiple index-terms
- **light AND therap***
- **Assumption:** the terms **therapy**, **therapies**, **therapeutic**, etc. are all equally predictive of relevance
- Do we need to modify the index to allow wildcard operators?

Wildcard operators

- During index construction, build a dictionary that maps prefixes to full terms.
- **therap*** : [therapy, therapies, therapeutic, ...]
- During query-processing, look up each prefix in the dictionary and expand the query using its full terms
- **light AND therap***

Wildcard operators

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- **therap*** : [therapy, therapies, therapeutic, ...]
- During query-processing, look up each prefix in the dictionary and expand the query using its full terms
- light **AND** therap*
- light **AND** (therapy **XX** therapies **XX** therapeutic **XX** ...)

Wildcard operators

- During index construction, build a dictionary that maps prefixes to full terms.
- **therap*** : [therapy, therapies, therapeutic, ...]
- During query-processing, look up each prefix in the dictionary and expand the query using its full terms
- **light AND therap***
- **light AND (therapy XX therapies XX therapeutic XX ...)**
- What boolean operator should **XX** be?

Wildcard operators

- Also can do this with suffixes:
 - ▶ ***eutic** : [hermeneutic, pharmaceutical, therapeutic, ...]
 - ▶ **light AND *eutic**
 - ▶ **light AND (hermeneutic OR pharmaceutical OR therapeutic)**

Wildcard operators

handle within-term wildcards (e.g., **therap*c**)

- Fetch list of full-terms for prefix **therap***
- Fetch list of full-terms for suffix ***c**
- Take the intersection

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